

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **test automation capabilities** for automotive camera systems, achieving **90% test execution coverage** against system-level requirements using Python scripting and platform standards.
- Executed comprehensive system testing for mechatronic ECU systems, ensuring **traceability of test specifications** to system requirements and validating fixes for **over 50 identified defects** during the project development lifecycle.
- Designed and implemented test methods for automotive software features, contributing to the on-time delivery of **system validation targets** by collaborating with System Engineers and Function Owners.
- Supported the integration and validation of Electronic Systems, assessing testability of architecture designs and planning test coverage for new ADAS features, **reducing integration issues by 15%**.
- Managed test environment readiness for Bench and HIL setups, ensuring consistent execution of planned testing phases and maintaining test results and logs throughout the project lifecycle, **improving test cycle efficiency by 20%**.
- Contributed to stress and robustness testing of end-user functions, identifying and accurately communicating system issues through detailed defect reports and status updates, **preventing critical failures in pre-production builds**.

SKILLS

Programming Languages – Python, C++, C, C#, Java, SQL.

Testing & Validation – System Testing, Test Automation, Test Specification, Traceability Analysis, Defect Reporting, Performance Testing, Stress Testing, Robustness Testing.

Automotive Systems – ADAS, Mechatronic ECU Systems, Automotive Camera Systems, Vehicle Environments.

Development Tools – Git, Jenkins, Jira, Confluence, Docker, AWS.

System Engineering – Requirements Analysis, Architecture Design, Systems Development Life Cycle.

Data Analysis & BI – Power BI, DAX, Tableau, R, Python (pandas, NumPy, scikit-learn).

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

ADAS Feature Validation Framework (Valeo Project)

- Developed a **Python-based test automation framework** for ADAS camera systems, enabling automated execution of functional and performance requirements and achieving **95% test coverage**.

- Implemented **traceability analysis** between system requirements and test specifications using Jira, ensuring comprehensive validation and reducing validation gaps by **25%**.
- Designed and executed stress and robustness tests for mechatronic ECU systems, identifying critical defects and contributing to the on-time delivery of system validation targets for new features.

Payments Operations Dashboard Enhancements (HCL Technologies)

- Enhanced a **payments operations dashboard** by developing new transaction monitoring features using **Java Spring Boot**, improving visibility into failed payment messages and reducing manual intervention time by **30%**.
- Automated data validation and reporting processes using **Python scripting** and SQL queries across PostgreSQL and Oracle databases, ensuring data accuracy for reconciliation workflows.
- Provided production support for the financial-services dashboard, responding to P1/P2 incidents and managing change-control updates in a GDPR-aware environment, sustaining **99.9% uptime**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)